Mind the Gap[1]

Measurement Dependence in the Brain Alignment of Jacobian-Transported Language Representations



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1 Research conducted at the Digital Minds Research Sprint, August 2026

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Abstract

Proposals that verbalizable language-model representations form a global workspace motivate comparison with human brain activity, but the result may depend on how the model representation and neural target are operationalized. We combine two independently implemented representational similarity analysis (RSA) studies that compare raw residual states with dense average-Jacobian transport, $J_l h_l$, for captioned Natural Scenes Dataset images. A whole-cortex searchlight study using Qwen3.5-4B and all-token mean pooling found no effect at layers 8 and 16 and small gains at layers 23 and 30 (J-minus-raw RSA = +.00165 and +.00146; Benjamini-Hochberg q = .04688). The gain was spatially near-uniform rather than concentrated in a higher-order network. A matched-region study using Qwen3-4B, last-token readout, and learned final RMS normalization instead found a reduction from mean RSA r = .09782 to .08558 across layers 18 and 27 and three higher-visual regions (mean difference = -.01224; 8/8 participants negative; exact p = .0078125). Its frozen prompt-by-depth extension produced 15 negative estimates among 18 new region-level contrasts. A no-final-normalization control reversed the layer-27 sign, providing a concrete bridge between pipelines without isolating one causal factor. Dense Jacobian transport therefore changes model-brain geometry, but not in a measurement-invariant direction. These studies do not test the formal sparse, non-negative J-space projection and do not establish a biological global workspace. Their joint contribution is a reproducible map of that inferential gap and a harmonized design for closing it.

1. Introduction

Representational similarity analysis (RSA) compares the geometry of model and neural response patterns without requiring matching feature dimensions [1]. It has revealed robust correspondence between language-model representations and high-level visual responses to natural scenes [3]. Yet a correlation between two dissimilarity matrices does not identify a common feature basis, algorithm, or biological mechanism. This distinction becomes critical when an interpretability construct is translated into a neuroscience experiment.

The Jacobian Lens maps an intermediate residual state h_l through an average layer-to-output Jacobian J_l and supports vocabulary-aligned analyses of representations described as verbalizable or workspace-like [4]. The strongest theoretical claims concern sparse, non-negative, interpretable structure. The two studies combined here test the simpler dense transport $J_l h_l$. Dense transport is scientifically useful as a controlled geometric perturbation, but it is not identical to the formal J-space projection and should not inherit its theoretical interpretation automatically.

We ask whether dense Jacobian transport changes caption-to-brain RSA consistently, where any change occurs across model depth and cortex, and which analysis choices can reverse its direction. The answer is not a single stronger-alignment score. Two pipelines produce different signs, while converging on depth dependence, weak spatial specificity, and a structural mismatch between caption-derived model inputs and image-evoked fMRI.

Our main contributions are: (1) a joint analysis of whole-cortex searchlight and targeted-ROI evidence; (2) participant-level randomization inference and frozen prompt, depth, spectrum, and normalization diagnostics; and (3) a falsifiable harmonization protocol separating dense transport from formal J-space and from biological-workspace claims.

2. Related Work

RSA was introduced as a common representational language across brains, behavior, and computational models [1]. NSD provides repeated 7 T fMRI measurements during natural-scene viewing [2], and Doerig et al. showed that LLM representations of scene semantics align with high-level visual cortex [3]. These results motivate an extension from ordinary embeddings to Jacobian-derived representations.

Gurnee et al. characterize verbalizable residual structure using average Jacobians and sparse non-negative decompositions [4], drawing an analogy to global-workspace theories in which information becomes broadly available for report [5]. This analogy predicts neither that every dense Jacobian transform should increase fMRI RSA nor that passive visual cortex should be its privileged neural target. Mahner et al. further show why scalar human-model alignment is insufficient: similar overall scores can conceal different semantic and visual dimensions [8]. Our gap is therefore operational: what survives when the same broad J-Lens idea is instantiated through different readouts and neural summaries?

3. Methods

3.1 Shared estimand and study architecture

Both studies used COCO descriptions of NSD images [2,7], causal Qwen decoder states, a matched raw residual h_l, and dense transported features $J_l h_l$. Within each analysis unit, model and fMRI response matrices were converted to correlation-distance representational dissimilarity matrices (RDMs). RSA was Pearson correlation between upper-triangular model and neural RDM entries. The paired effect was $\Delta = r(J_l h_l) - r(h_l)$; positive values indicate stronger RSA after transport. Participants, not image pairs or searchlights, were the unit of group inference.

Design choice	Study A: whole-cortex searchlight	Study B: targeted ROIs
Model / lens	Qwen3.5-4B; qwen-n1000 lens	Qwen3-4B; matching public lens
Model input	Caption text only	Separate captions; raw or imagery prefix
Readout	Mean of all valid tokens; J applied tokenwise	Last attended non-terminal-special token; captions averaged by image
Feature scaling	Native residual units; no feature normalization	Learned final RMSNorm after raw or transported state
Depth	Layers 8, 16, 23, 30	Primary 18/27; extension 9/18/27/34
Brain target	NSD whole-cortex searchlights	Bilateral ventral, lateral, parietal higher-visual ROIs

Design choice	Study A: whole-cortex searchlight	Study B: targeted ROIs
Stimulus sampling	8 participants; 8 disjoint 100-image samples each	515 images shared by all 8 participants; 2,578 captions
Group statistic	Whole-searchlight mean per participant	Mean layer-by-ROI contrast per participant

Table 1. The studies share the dense-transport contrast but differ on several coupled measurement choices. Their effect estimates should be compared, not pooled as exchangeable replications.

3.2 Study A: whole-cortex searchlight

Study A used Qwen3.5-4B with the public qwen-n1000 Jacobian Lens estimated from 1,000 WikiText examples. Layers 8, 16, 23, and 30 contributed raw and transported residuals; the final decoder block and an all-mpnet-base-v2 sentence representation were descriptive controls. Caption prompts were deterministic and contained no instruction or generated text. Features were mean-pooled over all valid non-padding positions, with J applied to each token before pooling. No per-vector L2 normalization or feature standardization was applied.

For each of eight participants, sessions 1-10 and three-repeat conditions yielded eight disjoint 100-image samples. Whole-cortex searchlights correlated local fMRI RDMs with all model RDMs. The participant summary averaged searchlight RSA within each matched sample and then across samples. Exact two-sided sign-flip tests enumerated all 2^8 sign assignments; Benjamini-Hochberg correction covered the four layer contrasts [6]. Regional analyses used normalized HCP-MMP1 parcel gains to test spatial concentration.

3.3 Study B: targeted ROIs and frozen extension

Study B analyzed 515 images shared across all eight NSD participants and 2,578 captions. Neural response patterns came from bilateral ventral, lateral, and parietal higher-visual ROIs. Captions were encoded separately; the last attended non-terminal-special-token state was retained and caption features were averaged within image. The baseline applied Qwen3-4B's learned final RMSNorm to h_l; the transported condition applied the same normalization to $J_l h_l$.

The preregistered raw-caption analysis averaged six contrasts per participant: layers 18 and 27 crossed with three ROIs. Its exact two-sided sign-flip test was the sole confirmatory test. After observing that result but before new extraction, we froze an extension adding raw-caption layers 9 and 34 and imagery-prompt layers 9, 18, 27, and 34. The imagery template was exactly: "Imagine the visual scene described by this caption: <caption>". Benjamini-Hochberg correction was applied once across the 18 new layer-by-ROI cells. The extension remains exploratory despite within-family correction.

3.4 Cross-study bridge diagnostics

Study B measured Jacobian singular spectra, raw-versus-transported model-RDM correlations, and a no-final-RMSNorm control at layers 18 and 27. These diagnostics distinguish hard rank loss from anisotropic reweighting and test whether final normalization can participate in a sign change. They do not form a causal harmonization study because the two pipelines also differ in model checkpoint, layer locations, pooling, stimulus sampling, and cortical summary.

4. Results

4.1 Study A: small late-layer gains without a focal network

Dense transport had negligible effects at layers 8 ($\Delta = -.00048$, 95% CI [-.00261, +.00166]) and 16 (+.00036 [-.00144, +.00216]). It increased whole-cortex mean RSA at layers 23 (+.00165 [+.00062, +.00268]) and 30 (+.00146 [+.00051, +.00241]); both corrected q values were .04688 (Figure 1A). At layer 23, raw and transported means were .02981 and .03146, a roughly 5% relative gain. The best transported feature exceeded the MPNet reference (.02674), but only modestly.

The gain was not localized to a candidate workspace system. After normalization for regional signal level, it was approximately proportional across cortex, and 94 of 180 parcels were individually significant. This broad profile is more consistent with a global change in representational geometry than with evidence for one higher-order cortical locus.

4.2 Study B: robust reduction and prompt-by-depth reversals

In the confirmatory aggregate, raw residuals reached mean RSA $r = .09782$ and transported residuals .08558. Every participant had a negative difference (mean $\Delta = -.01224$; SEM = .00141; Cohen's $d_z = -3.07$; exact two-sided $p = .0078125$; Figure 1B). Five of six layer-by-ROI cells were negative and FDR-significant; only layer-27 lateral cortex was positive and nonsignificant (+.00302).

The extension showed strong configuration dependence. Mean raw-caption effects across the three ROIs were -.07157, -.02134, -.00315, and -.02510 at layers 9, 18, 27, and 34. Imagery-prompt effects were -.02968, +.00430, -.01066, and -.02832 (Figure 1C). Fifteen of 18 new cells were negative; the three positive estimates all occurred under imagery at layer 18. Seventeen of 18 cells rejected after correction, with parietal imagery-layer-18 the only non-rejection (+.00286; adjusted p = .05469).

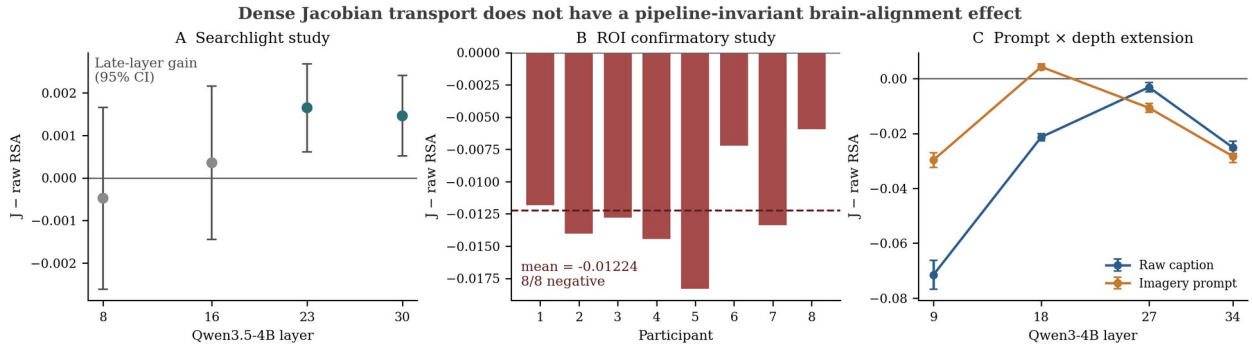


Figure 1. Dense Jacobian transport has no pipeline-invariant RSA direction. (A) Study A whole-cortex participant summaries, with 95% confidence intervals. Teal points survived four-test BH correction. (B) Study B preregistered aggregate differences for each participant; the dashed line is the participant mean. (C) Study B frozen extension, averaged across three ROIs; error bars are participant-level SEM. All panels show transported minus matched raw RSA.

4.3 Mind the gap: what reconciles and what remains unresolved

The studies do not estimate one common effect. Their shared manipulation is dense Jacobian transport, but Study A combines late Qwen3.5-4B layers, mean pooling, native feature units, and whole-cortex summaries, whereas Study B combines a different Qwen checkpoint, endpoint readout, final RMS normalization, and selected higher-visual ROIs. The sign discrepancy is therefore evidence of measurement dependence, not a failed reproduction of one fixed estimand.

The bridge diagnostics identify plausible contributors. All four 2,560 × 2,560 Jacobians were numerically full rank, but effective rank rose from 590 and 1,157 at layers 9 and 18 to 2,226 and 2,483 at layers 27 and 34. Raw-versus-transported model-RDM correlations likewise rose with depth (.697, .719, .848, .961 for raw captions), showing that later transport increasingly preserves baseline geometry. Without final RMSNorm, all layer-18 ROI effects remained negative (-.01557 to -.00695), whereas all layer-27 effects became positive (+.00201 to +.00910). This makes normalization-by-depth interaction a concrete candidate explanation, but not a uniquely identified cause.

5. Discussion and Limitations

The combined evidence rejects a simple claim that replacing raw embeddings with dense J-Lens embeddings generally improves brain alignment. It also rejects the opposite universal claim that transport necessarily harms alignment. What survives is narrower and more useful: transport reweights caption geometry strongly at some depths, weakly at others, and the measured neural effect depends on how that geometry is read out, normalized, and summarized.

This matters for interpretability and AI safety because a theoretically meaningful representation can acquire unwarranted biological authority through a convenient proxy. RSA supplies a test of shared relational structure, not a proof of shared semantic features, broadcasting, consciousness, or implementation. Study A's vocabulary diagnostics produced recognizable scene words, but those words largely paraphrased human captions already supplied to the model and were not used in the brain RDMs. A scalar RSA increase therefore cannot tell whether semantic, visual, lexical, or generic geometric factors drive the effect [8].

The brain-model input gap is structural. Participants viewed images; both language models received human-written captions, which already encode a human perceptual-to-semantic transformation. Moreover, NSD emphasizes passive visual responses, whereas workspace theories concern information availability for report, integration, and task-guided cognition [5]. More participants or stricter correction cannot repair this construct mismatch. The absence of a focal cortical gain in Study A further argues against presenting the current result as a neural workspace signature.

Limitations. Both studies use eight participants and one dataset. Their pipelines differ on several dimensions at once, so the present cross-study comparison cannot assign causality to pooling, normalization, model revision, layer depth, sampling, or spatial statistic. Study B's extension was frozen only after its primary result. Study

A's late-layer q values are just below .05 and the absolute gains are small. Neither study evaluates noise ceilings for every contrast, formal sparse J-space, encoding-model generalization, early visual controls within the same full design, or higher-level report tasks.

Future work. First, preregister a harmonized factorial replication using the same model, lens, stimuli, and brain summaries while crossing mean versus endpoint pooling, native versus final-RMS-normalized features, raw versus imagery prompting, and matched relative depths. Second, test the formal sparse, non-negative projection $\Pi_J(h_l)$, its residual complement $h_l - \Pi_J(h_l)$, the full residual, and variance- and dimension-matched random subspaces. Third, replace passive image viewing or caption proxies with language comprehension, deliberation, or explicit-report paradigms. Finally, combine RSA with encoding models and Mahner-style semantic-versus-visual dimensions [8]. This would ask what information aligns, not merely whether one scalar correlation changes.

6. Conclusion

Dense Jacobian transport produced a small late-layer increase in one whole-cortex pipeline and a robust decrease in a targeted-ROI pipeline, with prompt, depth, and normalization diagnostics showing reversible effects. The publishable result is therefore not that J-space is brain-like or brain-unlike. It is that the current neural conclusion is measurement-dependent.

Mind the Gap turns that discrepancy into a testable research program: align the estimand across pipelines, evaluate the formal sparse representation, and use neural tasks matched to verbalizable cognition. Until then, the data support a geometric sensitivity claim and a methodological warning—not a biological global-workspace claim.

Code and Data

Combined code, compact results, and reproducibility materials: https://github.com/frnkptrln/mind-the-gap-jlens-nsd. Study A's original whole-cortex implementation is pinned at https://github.com/lucas-nunn/natural-j-scenes/tree/8389a3897a29c043bf12462b804400e999baa330. The companion repository contains Study B's pinned configuration, preprocessing and RSA code, tests, manifests, participant-level tables, frozen analysis documents, and figure scripts. Natural Scenes Dataset: https://naturalscenesdataset.org/. Raw NSD data remain governed by NSD access terms and are not redistributed.

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Appendix A. Claim Boundary

Question	Current evidence	Status
Does dense $J_l h_l$ alter brain RSA?	Yes; direction depends on pipeline, depth, and prompt.	Supported
Is the gain localized to a workspace network?	Study A found approximately proportional cortical gain.	Not supported
Does formal sparse J-space align with brain activity?	$\Pi_J(h_l)$ was not tested.	Open
Does RSA establish a biological global workspace?	RSA does not identify mechanism, report availability, or consciousness.	Not tested

LLM Usage Statement

OpenAI Codex assisted with analysis code, artifact inspection, figure generation, and manuscript editing; Claude assisted with strategic critique. Numerical claims were checked against saved artifacts or public Study A summaries. The human researchers retain responsibility for the science; no AI system is an author or a source of neural data.